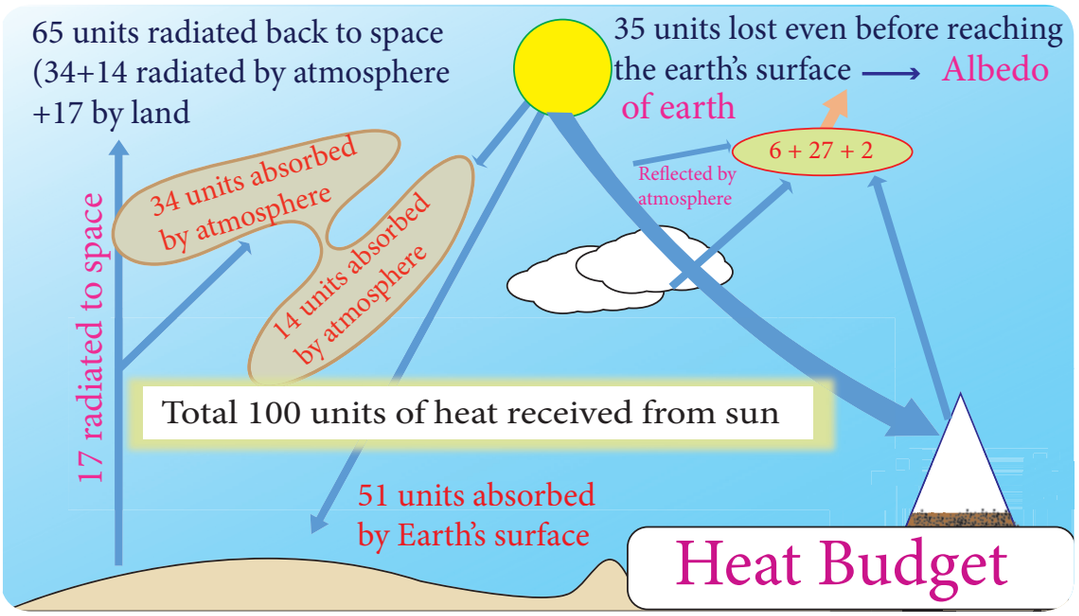


Figure 6.6 Heat Budget

back to space gradually. Therefore, 65% heat received from the sun is balanced by the 65% radiated by the earth. This balance between the incoming and the outgoing heat energy is called the **global heat energy balance**.

Distribution of Temperature

Distribution of temperature varies both horizontally and vertically. Let us study it under

a. Horizontal Distribution of Temperature
b. Vertical Distribution of Temperature

A) Horizontal Distribution of Temperature

Distribution of temperature across the latitudes over the surface of the earth is called horizontal distribution of temperature. On maps, the horizontal distribution of temperature is commonly shown by isotherms. **Isotherms** are line connecting points that have an equal temperature at mean sea level.



The average time taken by the solar radiation to reach the earth's surface is 8 minutes 20 seconds.

6.3.1 Factors Affecting the Horizontal Distribution of Temperature

The horizontal distribution of temperature on the earth's surface varies from place to place. Following are the factors affecting the horizontal distribution of temperature of the earth:

a. **Latitude:** The angle formed by the solar radiation to the ground is called 'angle of incidence'. The solar radiation passes vertically along the equator. The angle of incidence decreases from equator towards the poles. The area heated by the solar radiation increases towards the poles and therefore, temperature decreases from the equator to the poles.

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